- Safety-rated sensors monitor workspace
- Highest level of separation
- Suitable for inspection tasks

SAFETY MEASURES

- Safety controller monitors zones
- PLr d / Cat. 3 minimum
- IEC 60204-1 stop category 0 or 1

- Robot in reduced speed mode
- Operator in full control of motion
- Good for teaching and positioning

SAFETY MEASURES

- Speed limited (250 mm/s typical)
- Force-limited drive system
- Safety-rated speed monitoring

3 Speed & Separation Monitoring

MEDIUM

Robot speed adapts based on distance to human

KEY FEATURES

- Dynamic speed adjustment
- Separation distance monitored
- Robot slows as human approaches
- Stops if minimum distance violated
- Allows shared workspace with motion

SAFETY MEASURES

- Safety-rated speed monitoring
- Safety-rated distance monitoring
- ISO 13855 safety distance calc.
- T_{10} detection response time
- Protective separation distance Sp

4 Power & Force Limiting

HIGH

Robot limits contact forces to biomechanically safe levels

KEY FEATURES

- Inherent force/power limitation
- Direct contact permitted
- ISO/TS 15066 Annex A thresholds
- Body-region-specific limits
- Quasi-static and transient forces

SAFETY MEASURES

- Max. transient force (per body area)
- Max. quasi-static force (per body area)
- Pressure limits (clamping/impact)
- Energy transfer limits
- Rounded, padded robot surfaces

Risk Spectrum

Lower risk / More separation

Higher risk / Direct contact

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